

Project Title

Data Classification Using AI using K-Nearest Neighbors (KNN)

Objective

To build a machine learning classification model using the Iris dataset and classify flower species accurately.

Tools Used

- Python
- Google Colab
- Scikit-Learn

Dataset

Iris Dataset

Algorithm

K-Nearest Neighbors (KNN)

Steps Performed

1. Loaded Iris Dataset
2. Split data into training and testing sets
3. Applied StandardScaler
4. Trained KNN Model
5. Predicted test data
6. Evaluated model using Accuracy, Confusion Matrix and F1 Score

Results

- Accuracy: 100%
- Precision: 1.00
- Recall: 1.00
- F1 Score: 1.00

Conclusion

The KNN classification model successfully classified Iris flowers with 100% accuracy on the test dataset.

Screenshot

Project2.ipynb

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7

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accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

Accuracy: 1.0

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cm = confusion_matrix(y_test, y_pred)

print(cm)

[[10 0 0]
[0 9 0]
[0 0 11]]

9

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print(classification_report(y_test, y_pred))

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30